SECURITY LABS / 04

WINDOWS AUTHENTICATION / SPLUNK

Windows Authentication Detection with Splunk

From authentication records to a detection candidate

Synthetic-data detection analysis and expected results

Scope, dataset and technologies	02
Event fields and correlation keys	03
Rolling-window detection logic	04
Expected findings and scenario controls	05
Offline validation and Splunk execution	06
Investigation and evidence limits	07
Sources and evidence map	08

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SECURITY LABS / 04

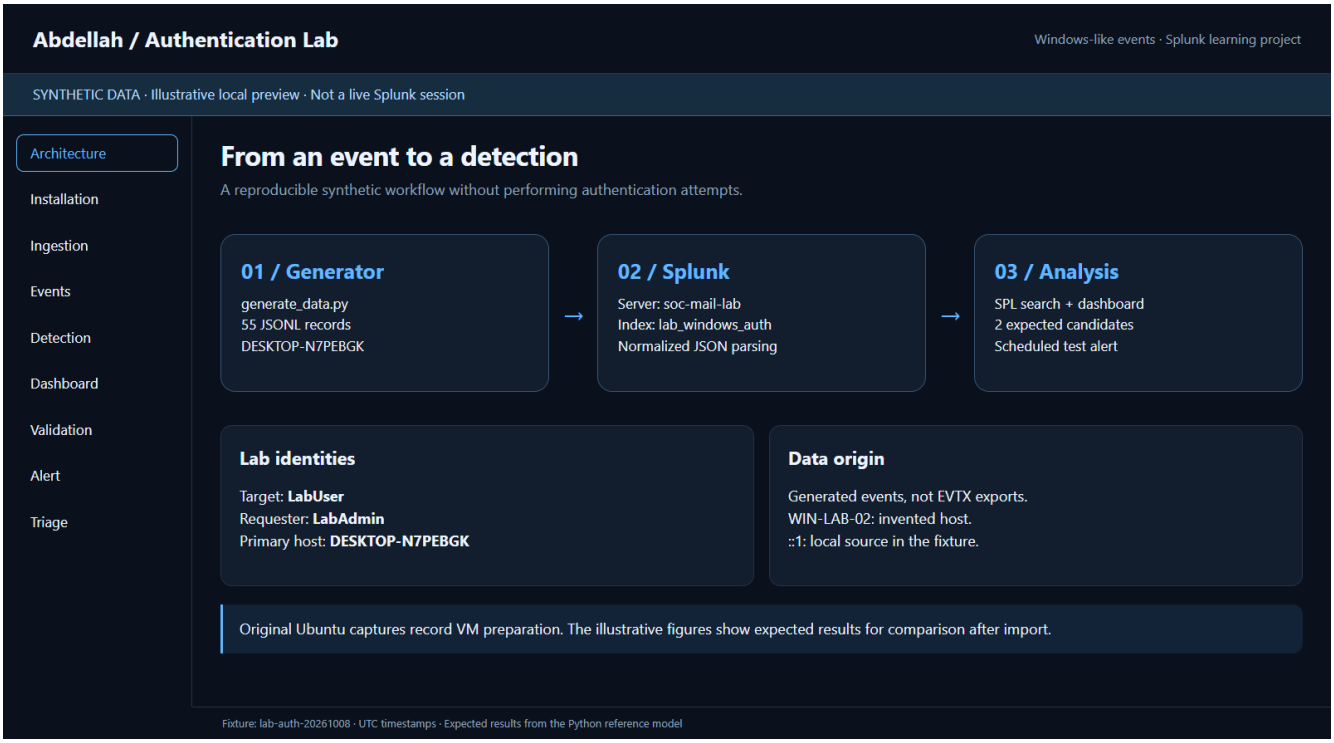
01 / Scope, dataset and technologies

The project defines an authentication detection exercise using Splunk searches and a generated Windows-like event dataset. It extends the earlier local Windows authentication investigation with a reproducible rule, dashboard and scheduled-alert configuration.

The historical fixture contains 55 records: 46 failed logons and nine successes across ten controlled scenarios. Lab names include DESKTOP-N7PEBGK, LabUser, LabAdmin and soc-mail-lab. WIN-LAB-02 is an invented control host.

Deliverables include JSONL/CSV data, eight SPL searches, an installable Splunk app, a Simple XML dashboard, a disabled scheduled alert and reproduction instructions. The record format is normalized JSON rather than an EVTX or native Windows XML export.

Verification status: the dataset, Python reference implementation, configuration structure and interactive preview have been checked offline. Execution of the SPL searches, dashboard and alert in Splunk remains pending.



Illustrative preview of expected results using the synthetic dataset.

SECURITY LABS / 04

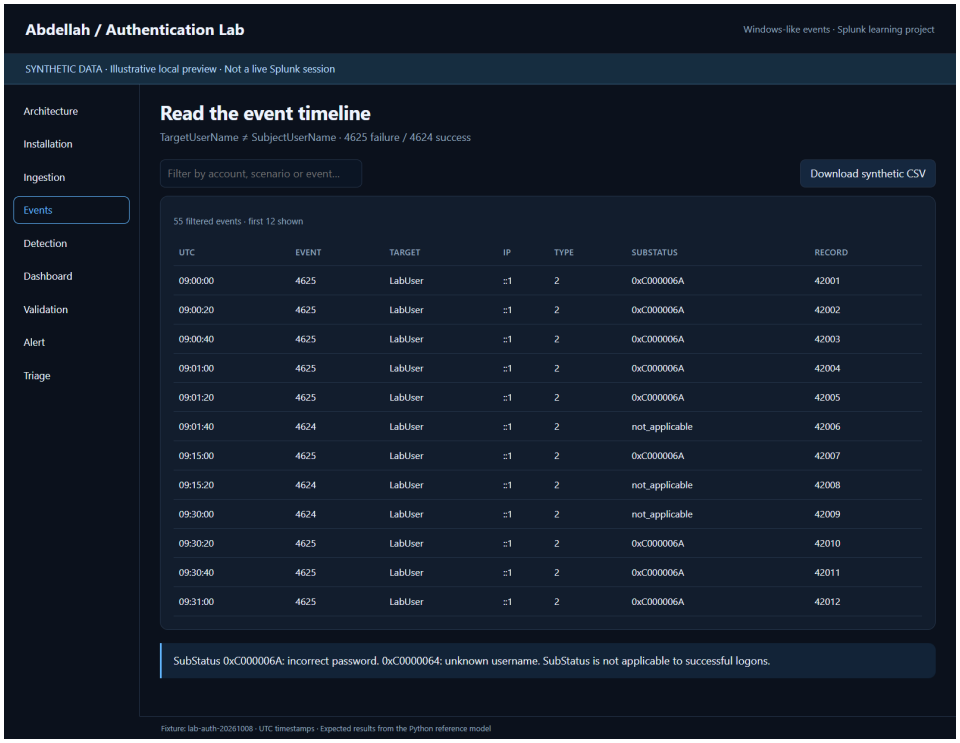
02 / Event fields and correlation keys

EventCode 4625 represents a failed authentication and 4624 a successful logon session. TargetUserName identifies the authentication target; SubjectUserName identifies the requester. Using the requester as the target would join the wrong identities.

The rule correlates run_id, Computer, TargetDomainName, TargetUserName, SourceIp and LogonType. Account and domain names are normalized to lower case. Record identity uses run_id, Computer and EventRecordID for deduplication.

SubStatus 0xC000006A represents an incorrect password in the fixture. 0xC0000064 represents an unknown username. LogonType 2 is interactive and 3 network. SourceIp is the project name for the normalized Windows IpAddress field.

The fixture uses ::1 for local test activity. The address of the Ubuntu Splunk server is unrelated to the authentication source. Missing correlation fields must be investigated rather than filled with invented values.



Illustrative preview of expected results using the synthetic dataset.

SECURITY LABS / 04

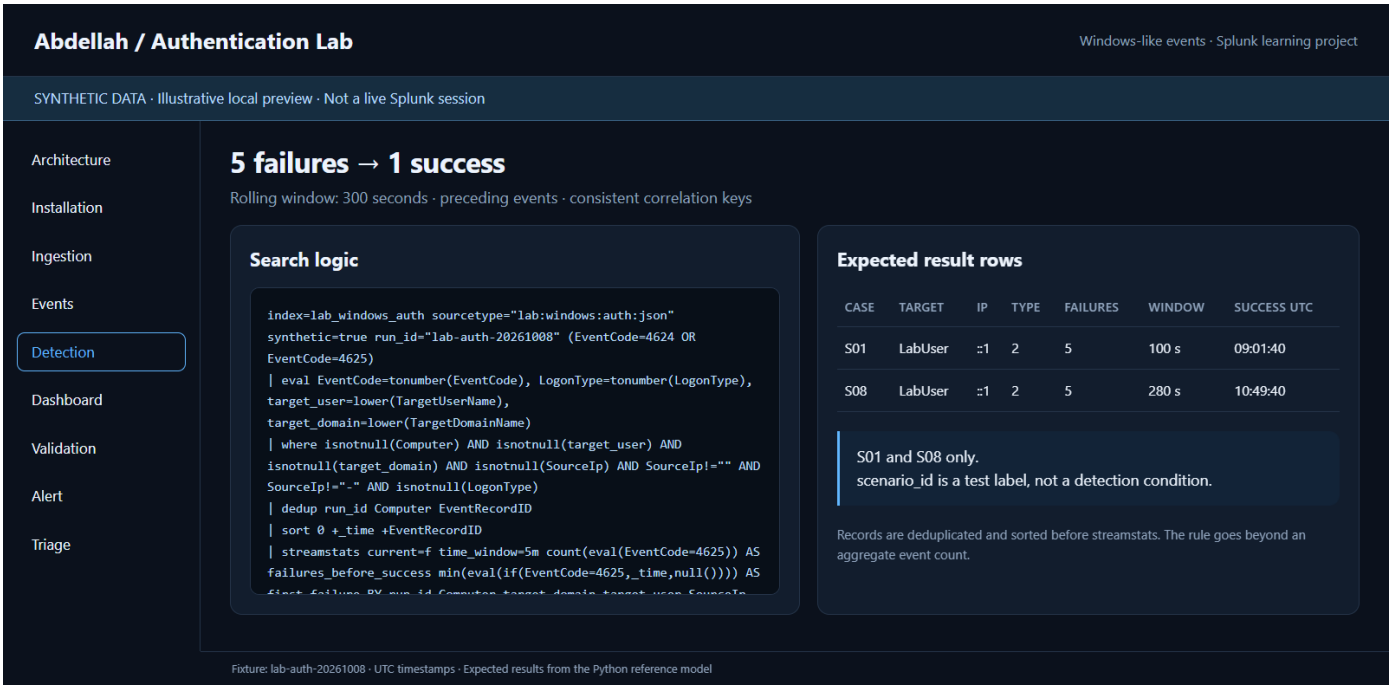
03 / Rolling-window detection logic

For each successful event, the search counts failures in the preceding five minutes for the same correlation key. The threshold is at least five failures. Events are deduplicated and sorted chronologically before streamstats calculates the window.

current=f excludes the current event from the running calculation. The success must follow the relevant failures; an earlier success does not establish a failed-then-success sequence. The full query is provided in queries/03-detection.spl.

scenario_id labels test data for comparison but is not a condition of the detection. Fixed five-minute buckets are avoided because a sequence can cross a bucket boundary.

This small-data design uses sort 0. Production deployment requires separate performance testing, delay handling, threshold tuning and detection coverage assessment.



Illustrative preview of expected results using the synthetic dataset.

SECURITY LABS / 04

04 / Expected findings and scenario controls

The reference result contains two candidates. S01 has five failures followed by a success at 09:01:40 UTC, 100 seconds after the first failure. S08 has five failures followed by success at 10:49:40 UTC over 280 seconds. Both use LabUser, DESKTOP-N7PEBGK, ::1 and LogonType 2.

Eight negative cases check a single mistyped password, a success before the failures, another target account, failures outside the window, another source address, another logon type, an unknown username without a success, and another computer.

Expected result IDs and counts are in data/expected-detections.csv and report/test-matrix.csv. These files contain reference output, not results exported from Splunk.

The two matching patterns are intentionally present in generated data. They are candidates for investigation and do not demonstrate credential theft or compromise.



Illustrative preview of expected results using the synthetic dataset.

SECURITY LABS / 04

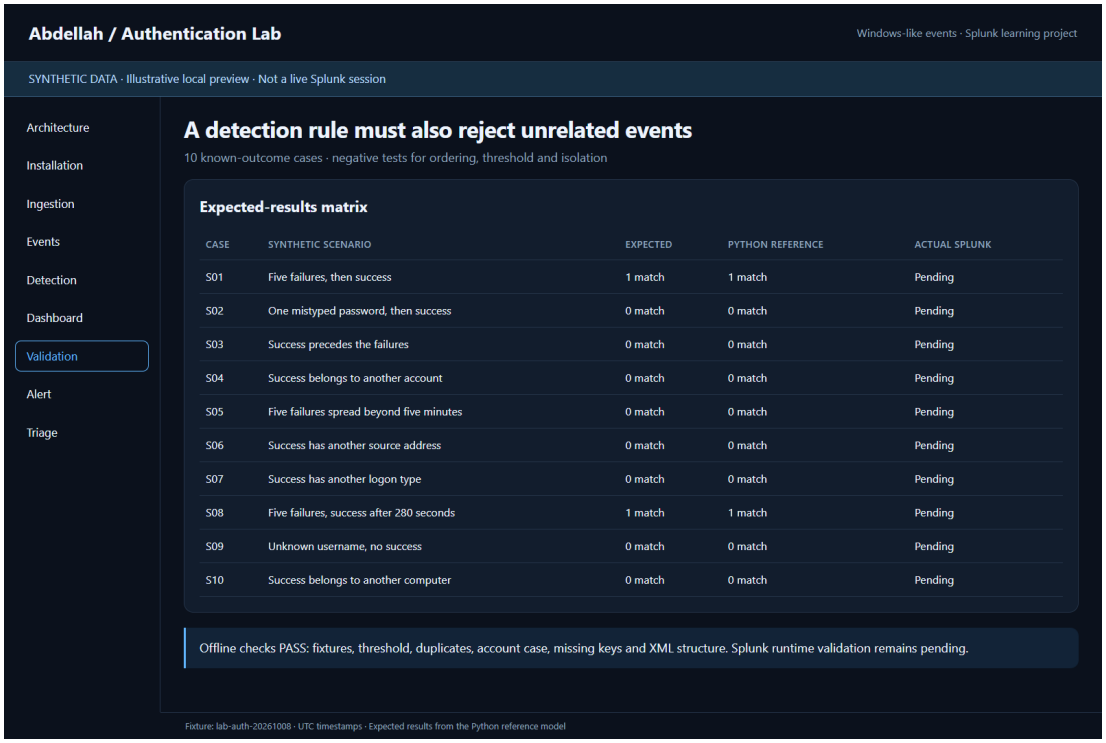
05 / Offline validation and Splunk execution

Offline checks passed for all ten scenarios. Additional checks cover duplicate records, a four-failure sequence, account-name case, a missing source field, a changed target domain and XML/configuration structure. The live-fixture generator also produces a valid six-event synthetic test.

Preview checks passed for filtering, CSV download and mobile layout. Screenshots of that preview are illustrative figures. Original Ubuntu terminal captures document the VM prerequisites only.

To validate in Splunk, install the app, search over All time, confirm 55 events and compare both detection IDs with expected output. Then enable the live alert, generate a current-time fixture and record the scheduled job and Triggered Alerts result.

The alert runs each minute with a one-minute event-time delay and suppresses repeated detection IDs for ten minutes. The schedule assumes limited ingestion delay and must be adjusted for late-arriving real telemetry. No external notification or containment action is configured.



Illustrative preview of expected results using the synthetic dataset.

SECURITY LABS / 04

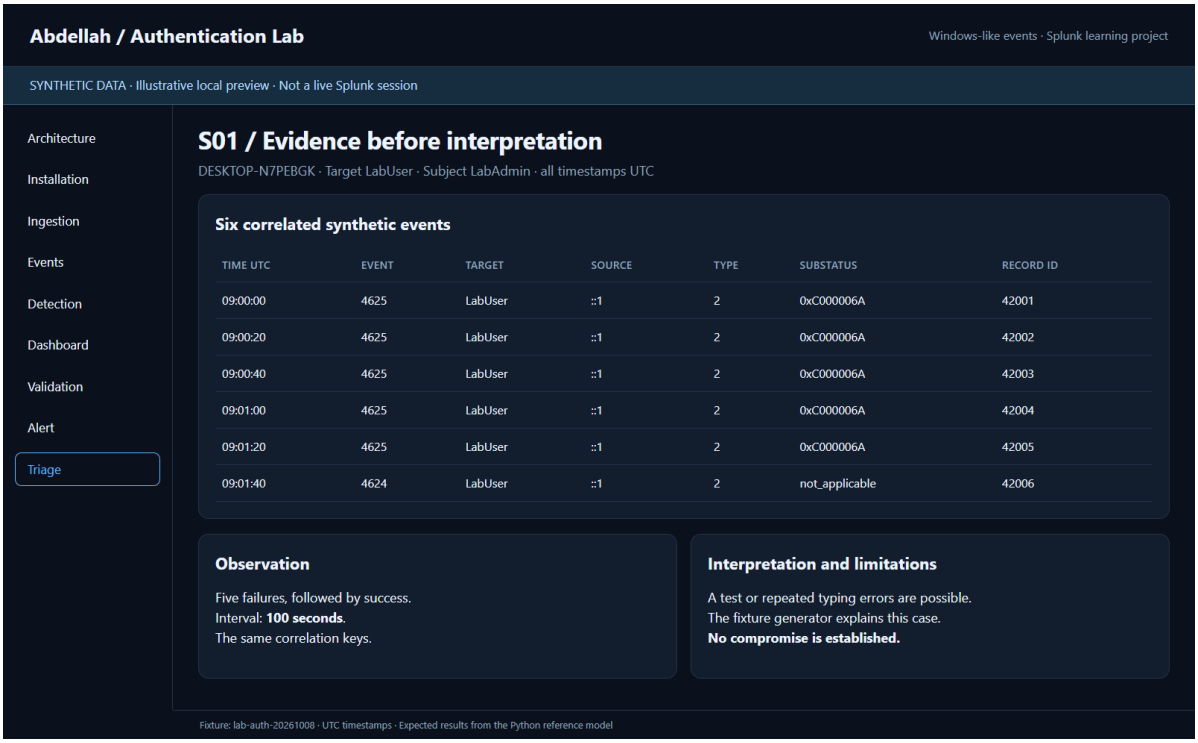
06 / Investigation and evidence limits

Observed in the fixture: five errors precede one accepted authentication for the same key. Possible explanations in real telemetry include repeated manual mistakes, administrative tests or unauthorized attempts. The logon result alone does not identify the person or intent.

Follow-up checks should confirm the operator and test context, preserve original events, inspect process and session details, review post-logon activity, search other accounts or hosts and assess audit completeness.

Strict source and logon-type correlation can miss activity that changes either key. Missing fields are excluded. Repeated successes can create separate candidates. Exact boundary behavior at 300 seconds is a separate Splunk check.

This project does not assert a production incident, EDR response or completed Splunk runtime validation. The optional export script reads selected lab-account events without creating users, changing audit policy or generating authentication attempts.



Illustrative preview of expected results using the synthetic dataset.

SECURITY LABS / 04

07 / Sources and evidence map

data/authentication.jsonl: generated historical records, run lab-auth-20261008. data/manifest.json: record totals and known-outcome scenarios. data/expected-detections.csv: expected candidates from the reference implementation.

queries/: copyable searches. app/abdellah_auth_lab/: parsing, input, index, saved searches and dashboard. verification.json and preview/verification.json: offline test results. report/actual-validation-template.md: future runtime evidence record.

screenshots/: original Ubuntu prerequisites and clearly labeled illustrative previews. evidence/actual-splunk/ is reserved for captures and exports from the actual Splunk instance.

Splunk Enterprise download

Linux installation

Run Splunk as a non-root user

SPL streamstats reference

props.conf reference

savedsearches.conf reference

Microsoft: event 4624

Microsoft: event 4625